PRATHEEP S

Chennai, Tamil Nadu, India | +91 7358054637 | pratheepsara2000@gmail.com | github.com/pratheep0707

PROFESSIONAL SUMMARY

Second-year B.Tech Computer Science student at VIT Chennai seeking an analytics internship. Built a Power BI dashboard on 1,470 employee records that estimated an $11.27M attrition cost and identified the main drivers of turnover, plus a Python machine learning system for soil erosion risk. Skilled in SQL, Python, Power BI, DAX, Excel and data visualization.

EDUCATION

Vellore Institute of Technology, Chennai, Tamil Nadu, India

Bachelor of Technology in Computer Science and Engineering | Expected Graduation: 2029

Central Board of Secondary Education (CBSE)

Class XII: 92.0% | Class X: 93.4%

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, C, C++
- **Data Analysis:** Exploratory Data Analysis, Data Cleaning, Data Preprocessing, Data Transformation, Feature Engineering, Statistics, Data Visualization, Dashboard Development
- **Tools:** Power BI, DAX, Power Query, Advanced Excel, Jupyter Notebook, Google Colab, Anaconda, Git, GitHub
- **Databases:** SQL, Oracle Database
- **Computer Science:** Data Structures and Algorithms, Operating Systems, Object-Oriented Programming

PROJECTS & EXPERIENCE

HR Workforce Dynamics and Attrition Cost Analysis | Power BI, DAX, Excel, Power Query

- Built a Power BI dashboard on 1,470 employee records that estimates a total replacement cost of $11.27M, using SHRM industry benchmarks and a star schema with a five-tier seniority cost table (0.50x to 1.50x annual salary).
- Wrote DAX measures showing a 16.12% baseline attrition rate, with 42% of exits within two years and 30.5% attrition among overtime employees against a 10.4% baseline.
- Created a flight-risk metric from promotion delays, satisfaction scores, overtime and pay tier to give HR a retention watchlist.
- Added departmental and overtime slicers and conditional formatting that show $4.63M of cost concentrated in Sales Executives.

Soil Erosion Prediction and Risk Assessment System | Python, Scikit-learn, GeoPandas

- Built a machine learning pipeline that predicts annual soil loss (t/ha/yr) using the RUSLE model and supervised regression on cleaned soil, rainfall, land cover and slope data.
- Trained separate urban and rural models and saved feature importance outputs.
- Produced GeoJSON and PNG risk maps with five severity levels and deployed a browser-based tool for region-wise erosion risk.